Kothapalli Akhil

ee24btech11035@iith.ac.in | kothapalliakhil1086@gmail.com | +91 9704960508 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.31
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	98.00%
Sri Sai Ram High School	Class X (Telangana State Board of Secondary Education)	2022	10.00%

SKILLS

Programming Languages : Embedded C, C++, Python.

Tools : Matlab | Intel Quartus Prime Lite | Icarus Verilog | GTKWave | KiCAD | Altium Designer | LTSpice.

PROJECTS

5 Stage Pipelined RISC-V Processor

- Implemented a **32-bit RISC-V processor with a 5-stage IF-ID-EX-MEM-WB pipeline**, integrating instruction fetch, register file, ALU, control logic, and instruction,data memories.
- Developed **pipeline registers and hazard-handling logic**, including operand forwarding, pipeline stalls and flushing to resolve data and control hazards.
- Implemented instruction decoding and datapath/control logic for **R, I, S and B Type instructions**, and verified processor operation through custom testbenches.

Embedded Scientific Calculator

- Developed a scientific calculator on an AVR microcontroller with **16×2 LCD interfacing** and **matrix keypad input**.
- Implemented arithmetic and scientific functions, including trigonometric operations, using **Taylor series-based numerical approximations**.
- Designed an expression parser for **real-time evaluation** and optimized firmware for responsive computation and display.

R2SDF Pipelined FFT hardware accelerator

- Implemented **Decimation in Frequency** (DIF) 8-point FFT accelerator in Verilog.
- Using a streaming **Radix-2 Single Delay Feedback** (R2SDF) architecture with fixed-point arithmetic.
- Implemented three **pipelined butterfly stages** with delay-feedback registers for streaming operation.
- Synthesized the design in Intel Quartus Prime Lite, achieving **235 MHz Fmax** with **242 ALMs and 1,092 registers**.

Q-Learning Algorithm Implementation

- Implemented a Q-Learning algorithm to learn an optimal control policy for a **battery management Markov Decision Process (MDP)**.
- Implemented -greedy exploration and iterative Bellman-based Q-value updates to learn the optimal policy without prior knowledge of the environment.
- Analyzed the effect of learning rate, discount factor, and exploration probability on convergence; validated the learned policy against an exact Value Iteration solution, achieving >90% policy agreement across discount factors.

RELEVANT COURSES

Computer Science : Introduction to Programming | Data Structures and Algorithms | Machine Learning | DBMS.

Electronics Courses & Labs : Digital Systems Theory & Lab | Electronic Devices & Circuits Lab | FPGA Lab | PCB Design and Layout for Electronic Systems | Semiconductor Device Fundamentals | Scientific Programming for Electrical Engineering | Probability & Random Processes.

POSITIONS OF RESPONSIBILITY

Head of Events Domain | Elan & Vision, Annual Techno-Cultural Fest, IIT Hyderabad

- Managing a team of 15 Coordinators, leading the planning and execution of workshops, technical and cultural events, while overseeing logistics, budgeting, scheduling, and cross-domain coordination.

Internship Coordinator | Office of Career Services of IIT Hyderabad

- Coordinating in outreach activities and assisting in organizing the institute's internship recruitment drive.

Core Member | Elektronica, Electronics & Signal Processing Club of IIT Hyderabad.

- Gained Hands-on Experience on **PCB design**. Participated in various competitions, Coordinated in conducting technical sessions for juniors.

EXTRACURRICULAR

Won **3 medals** in various inter-hostel, inter-departmental filmmaking competitions as a Cinematographer | Participating in social-cause walkathons and Clean India initiatives as an NSS volunteer.